

Design and Manufacturing of Mach-Zehnder interferometers in SOI with varying arm length and refractive index.

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Abstract

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1. Introduction

Empty (I apologize for the incomplete report)

2. Theory

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3. Methods and Results

The waveguide geometry (height, width), polarization.

Strip Waveguide, 220 nm x 500 nm, TE polarization

The simulated waveguide mode profile

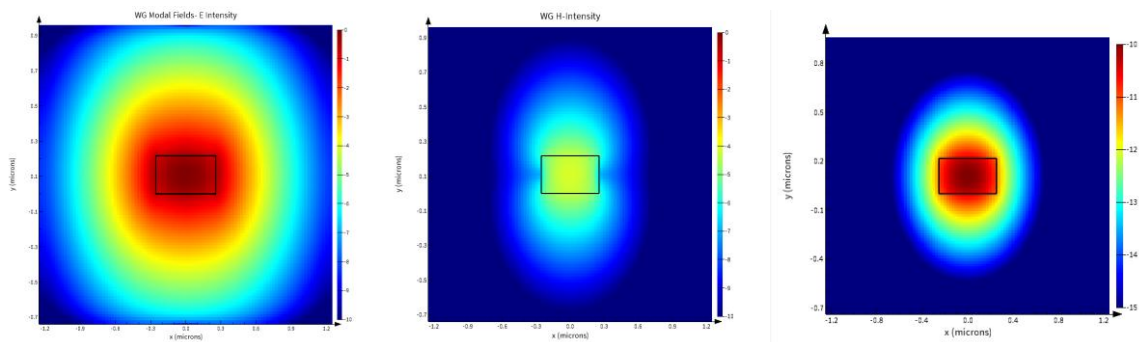


Figure 1. 1st mode TE polarization. The images were simulated in Lumerical Mode and show distribution of E (left) and H (middle) intensities inside the waveguide. (right) The energy density inside of the waveguide.

From the simulated mode profiles shown in Figure 1, it can be seen that the field intensity remains confined within the waveguide boundaries (Figure 1, left and middle).

The image in Figure 1 on the right shows that the energy is contained mostly within the core of the waveguide.

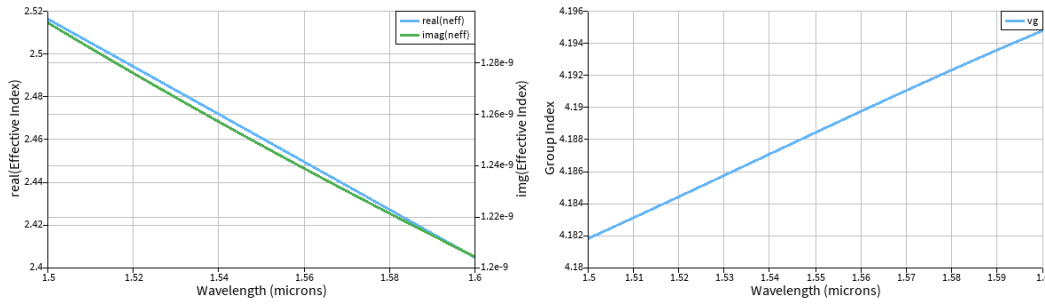


Figure 2. Left- effective refractive index of the WG simulated in Lumerical Mode. Right- group index of the WG simulated in Lumerical Mode.

To calculate to the compact model for the waveguide, the neff and ng data of the WG simulated in Lumerical mode were exported and analyzed in Python.

Compact model for the waveguide:

$$neff = 2.447 - 1.134 * (\lambda - \lambda_0) - 0.041 * (\lambda - \lambda_0)^2$$

The equation below describes the transfer function of the unbalanced Mach-Zehnder Interferometer

$$\frac{I_o}{I_i} = \frac{1}{2} [1 + \cos(\beta_1 L_1 - \beta_2 L_2)] \quad , \text{ where } \beta_{1,2} = \frac{2\pi \left(n + \frac{dn}{dt} \Delta T_{1,2} \right)}{\lambda} \quad \text{is a propagation constant.}$$

By changing the parameters of the propagation constant, e.g. temperature, it is possible to manipulate the phase, and thus, the intensity of the output. A typical transfer function of an unbalanced MZI is shown in Figure 3 below.

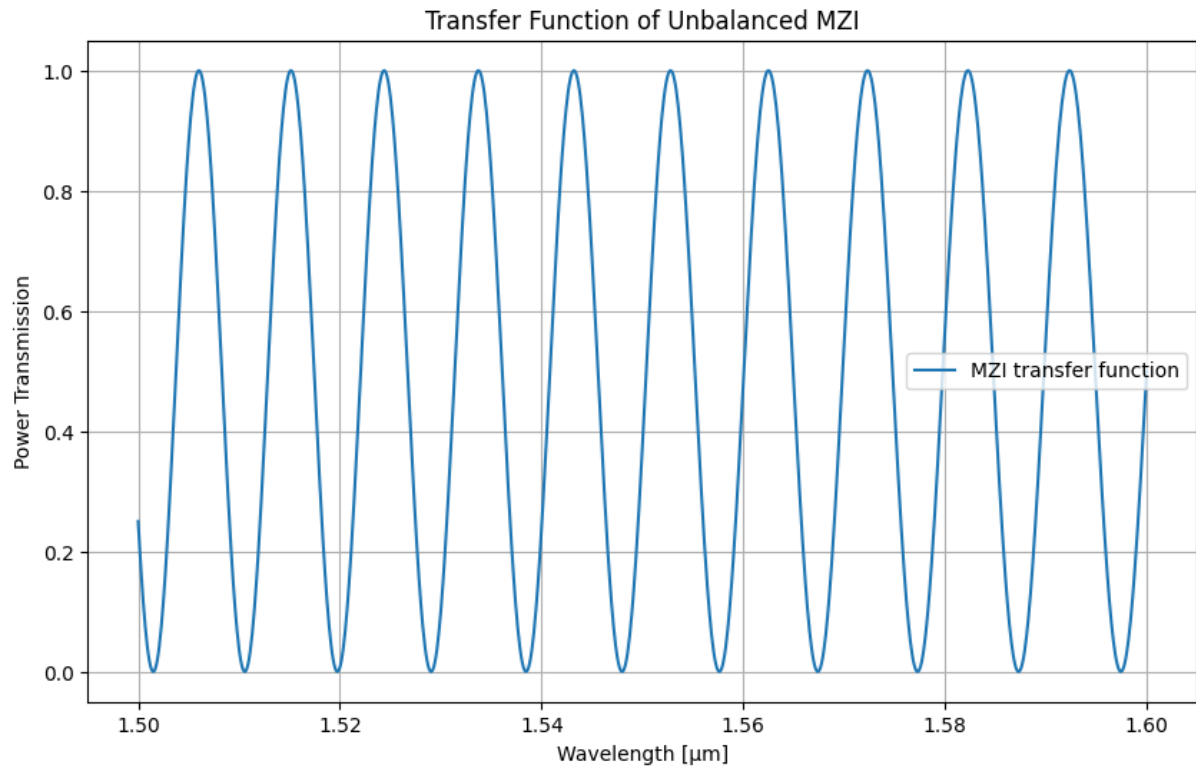


Figure 3. A typical transfer function of an unbalanced MZI.

In this project, I investigated how changing the path difference between the arms of interferometer and changing n_{eff} in one of the arms of an imbalanced MZI affects the FSR (Figure 4).

Path difference

The FSR around the central wavelength 1.55 μm can be calculated as follows:

$$FSR = \frac{\lambda^2}{n_g * dL}$$

where $dL_{\mu m}$ – path difference, n_g – group index at 1.55 μm, λ – 1.55 μm wavelength.

The table 1 shows FSR the values calculated based on the parameters of the compact model of the WG and the values simulated in Lumerical Interconnect, which show a good match.

Table 1

	$dL, \mu m$	FSR compact model, nm	FSR Lumerical Interconnect, nm
MZI1	269	2.126	2.1
MZI2	347	1.648	1.65

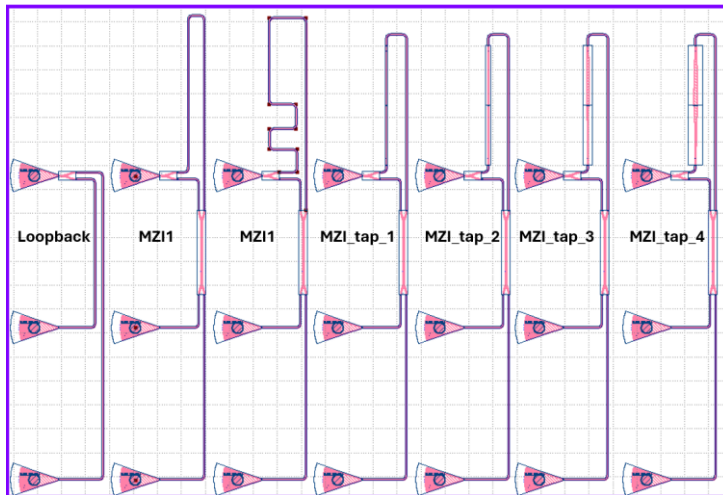


Figure 4. Loopback circuit to account for coupling loss. MZI 1 and 2 are used to investigate effect of path difference. MZI tap 1 to 4 to investigate effect of refractive index differences.

Refractive index change

To simulate the change of the refractive index in one of the arms of the interferometer, I used tapered waveguides and calculated the change of FSR in an unbalanced MZI.

First, 4 designs (Figure 4) of an unbalanced MZI were designed in KLayout with varying width of tapers from 500 nm to 3000 nm and constant height of 220 nm. The difference of the path length were kept constant and equals 237.52 nm. The first MZI in fig starting from the left has 2 tapers, which have width₁=500 nm, width₂ =500 nm. The tapers connected in the way that larger side are connected to avoid width mismatch. The second MZI has tapers with width₁=500 nm , width₂ =1000 nm. The third MZI has tapers with width₁=500 nm, width₂ =2000 nm. The fourth MZI has tapers with width₁=500 nm, width₂ =3000 nm.

To analyze the transmission spectrum I ran simulations of the MZIs designed in KLayout in Lumerical Interconnect. Unfortunately, the simulation showed that FSR does not change as a function of taper width (see Figure 5). Most probably due to the following error: *MZIneff3::ebeam_bdc_te1550_4::SPAR_1: Warning: s-parameters failed passivity test, passivity will be enforced. MZIneff3::ebeam_taper_te1550_8::SPAR_1: Error loading file:*

'/Users/lukasc2/.config/Lumerical/Custom/EBeam/source_data/ebeam_taper_te1550/w1=0.5um_w2=3um_length=20um.dat'

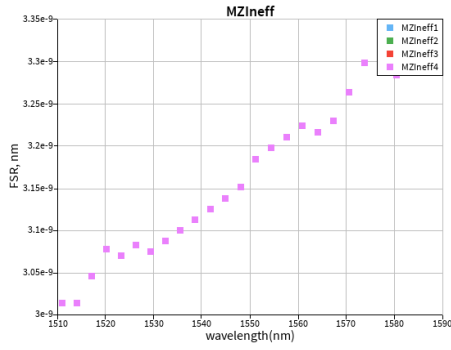


Figure 5. All calculated FSRs from different MZIs coincide.

To further analyze how taper width changes FSR of an unbalanced MZI, I simulated a cross-section of a waveguide in Lumerical Mode and simulated effective refracted index and group delay refractive index. Then, I ran a script in Python to iterate through all possible width of the tapered waveguide ranging from 500 nm to 3000 nm and to calculate its parameters.

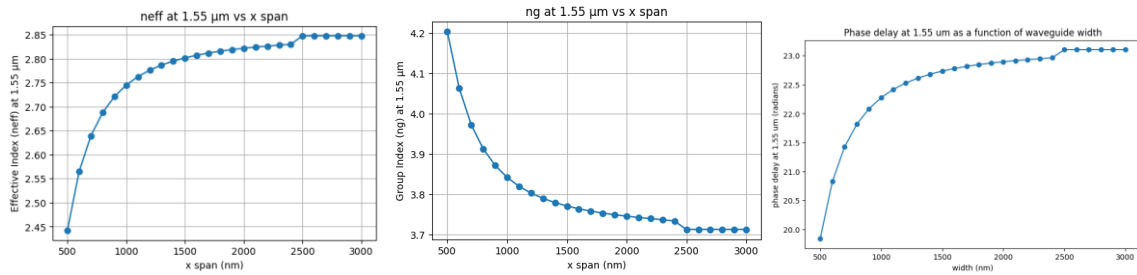


Figure 6. neff and ng as a function of the width of the tapered waveguide.

Figure 6 (left) shows that at larger width of the waveguide, the effective refractive index increases. That means that most of the propagating mode is in the high refractive index silicon core and the phase delay over the length of the taper decreases as well as the group delay.

The phase delay graph (Figure 6-left) shows how the phase is delayed in the taper. The trend is not linear, thus simple averaging over the length of the waveguide will not produce accurate results. It is required to integrate the accumulation of the phase at along the length of the taper to calculate the correct phase delay for each wavelength.

Furthermore, to calculate the FSR for each of the tapered MZI, the phase delay was integrated along each arm of the interferometer and the MZI transfer function is calculated as a function of the phase difference (Figure 7).

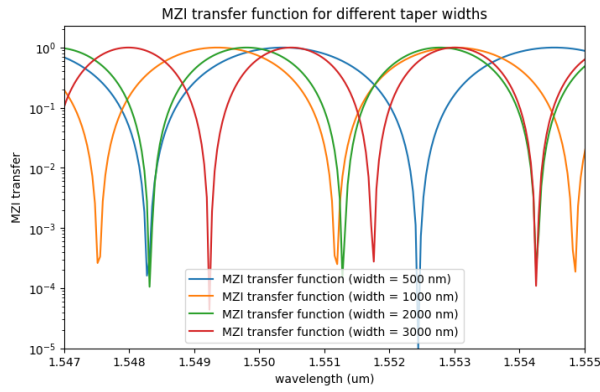


Figure 7. FSR of the tapered MZI at different widths of the taper

Table 2 shows the calculated values of FSR for different widths of the taper.

Table 2

Corresponding MZI	Taper width, nm	FSR, nm
MZI_tap_1	500	4.149
MZI_tap_2	1000	3.643
MZI_tap_3	2000	2.976
MZI_tap_4	3000	2.501

The simulated results for the MZIs containing different tapers show that the larger the taper width, the smaller the FSR. It can also be seen that wider tapers produce greater accumulated phase delay.

Manufacturing Challenges

To take into account Manufacturing Challenges I performed Corner Analysis for a range of width and heights of the waveguide. The Corner Analysis allows to estimate the range of expected variation of MZI parameters. Figure 8 shows n_{eff} and n_g as a function of wavelength for different width and heights of the waveguide.

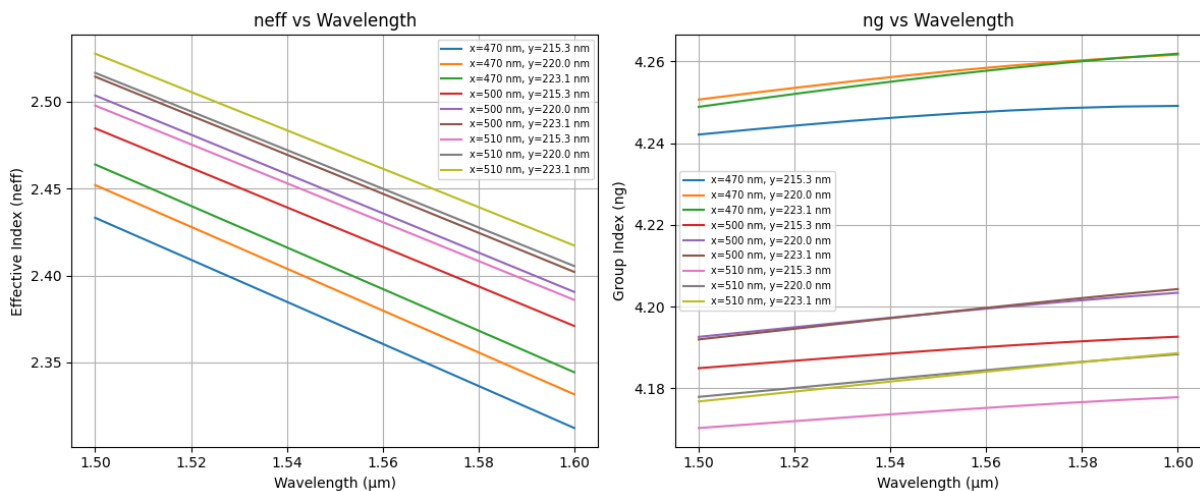


Figure 8. (left) Effective refractive index of the waveguide. (right) Group index of the waveguide.

Furthermore, the min and max FSR were calculated for $dL=347$ μm : $\text{FSR}_{\text{max}} = 1.65$, $\text{FSR}_{\text{min}} = 1.62$.

Data analysis

After the fabrication of the designed PICs, the data was analyzed and plotted. First, the coupler spectral response (loopback) from the measured data was fitted with a curve and used in all the following calculations to compensate for the coupling losses (Figure 9).

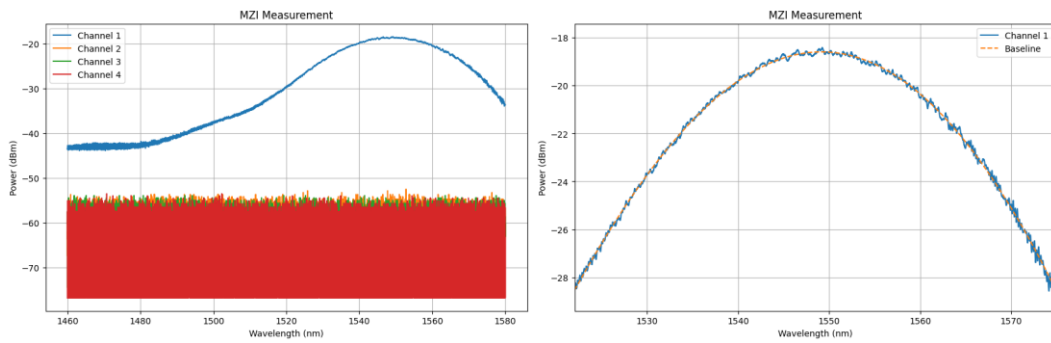


Figure 9. Signal from the loopback to calibrate for the insertion loss. B. fitted polynomial

As an example, the data from the MZI2 are analyzed in detail. The initial and corrected for the baseline spectrum of MZI2 are shown in Figure 10.

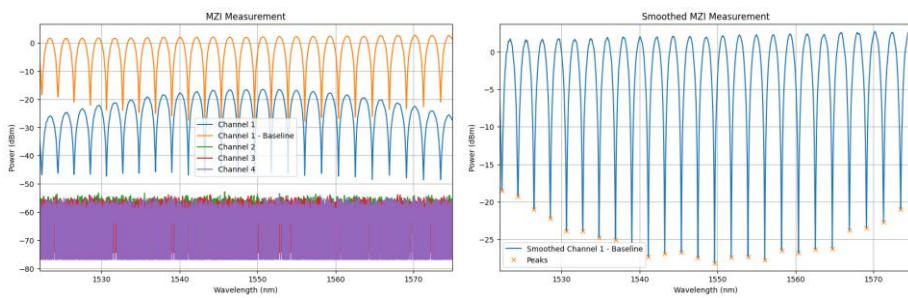


Figure 10. (left) raw and corrected for the baseline spectrum of the MZI2. (right) smoothed spectral data of MZI2 and found peaks.

To extract FSR from the measured data, one of the approaches is to find peaks corresponding to the dips in the spectrum and calculate the difference between the neighboring peaks (see Figure 11 - left). Furthermore, the group index is estimated (Figure 11 - right).

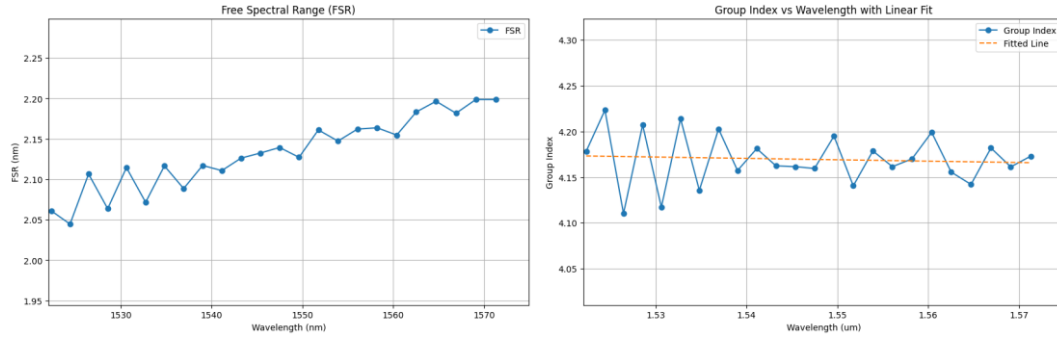


Figure 11. (left) FSR of MZI2 based on found peaks in the spectrum. (right) Group index from the found peaks.

To better estimate compact model of the effective refractive index of the measured MZI, a fitting and correlation analysis of the transfer function can be performed (see Figure 12).

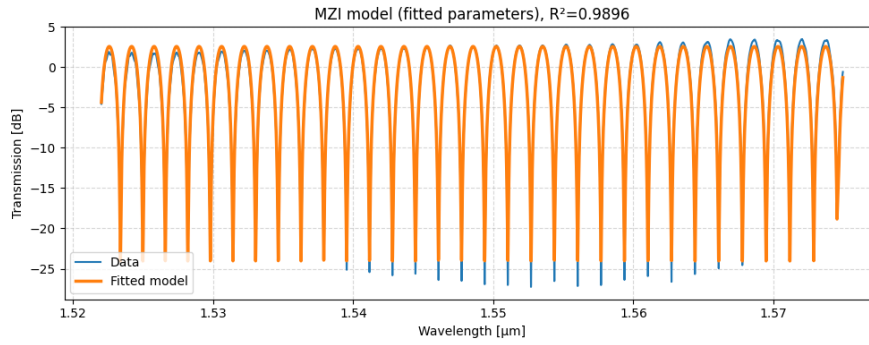


Figure 12. Correlation and fitting of the transfer function into the measured data.

From the fitted model the coefficients of the compact model of the effective refractive are calculated (Figure 13):

$$neff_{MZI2} = 2.446 - 1.906 * (\lambda - \lambda_0) - 2.7 * 10^{-4} * (\lambda - \lambda_0)^2$$

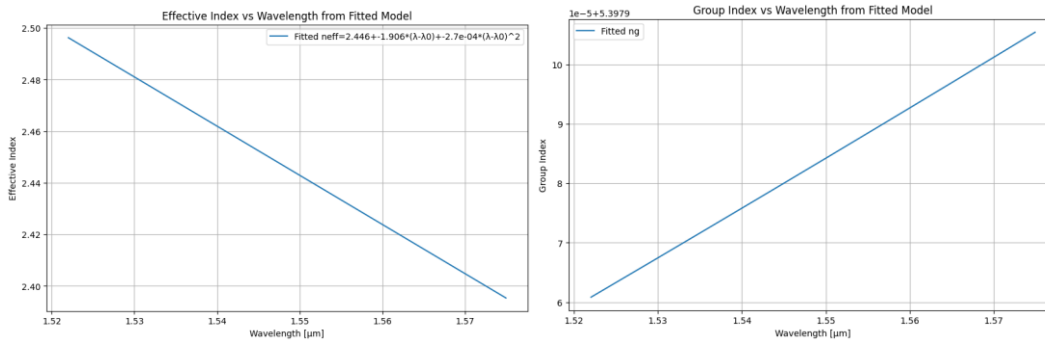


Figure 13. Effective refractive index (left) and group index (right) of the measured MZI2 based on the fitted model.

The table 3 shows the FSR values for all MZI present in the design. The corner analysis was performed for the first 2 MZIs. It shows that the FSR values are within the range of corner values, albeit very close to the limits.

The FSR Simulation and FSR measured value show quite significant discrepancy for tapered MZI. The discrepancy can be attributed to the mistakes made in simulations, which require revision.

Table 3

MZI Name	FSR Simulation, nm	FSR Measured, nm	FSR [min, max]
MZI1	2.126	2.139	[2.09, 2.14]
MZI2	1.648	1.654	[1.62, 1.65]
MZI_tap1	4.149	2.422	
MZI_tap2	3.643	2.394	
MZI_tap3	2.976	2.513	
MZI_tap4	2.501	2.523	

Table 4 contains measured n_{eff} , n_g , and FSR values at 1550 nm wavelength for all MZIs present in the design.

Table 4

MZI Name	n_{eff}	n_g	FSR, nm
MZI1	2.4441	4.1742	2.139
MZI2	2.4449	4.1801	1.654
MZI_tap1	2.4434	4.1770	2.422
MZI_tap2	2.4423	4.2260	2.394
MZI_tap3	2.4463	4.0260	2.513
MZI_tap4	2.4479	4.0089	2.523

References.